TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

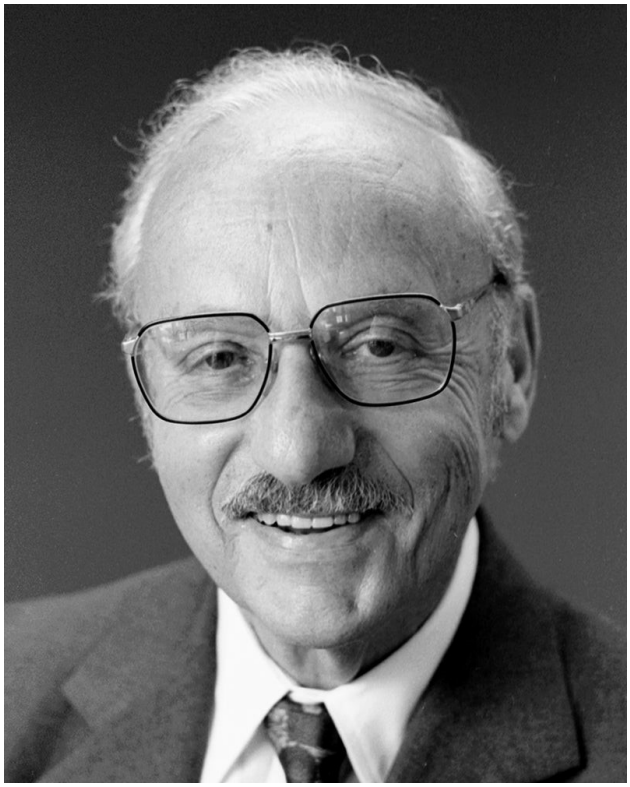


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

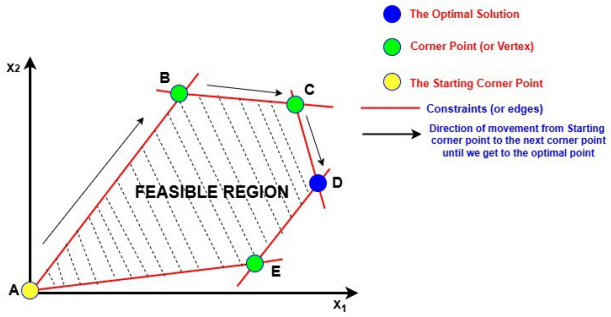


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Problema 8
- Amount of decision variables: 4
- Amount of restrictions: 2
- Total Amount of variables (decision + slack): 6
- Type of problem: Maximization

Equations

$$\text{Max } Z = 36\,x_1 + 30\,x_2 - 3\,x_3 - 4\,x_4$$

subject to:

$$x_1 + x_2 - x_3 \leq 5$$
$$6\,x_1 + 5\,x_2 - x_4 \leq 10$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Initial Table

Z	x_1	x_2	x_3	x_4	s_1	s_2	b
1.0000	-36.0000	-30.0000	3.0000	4.0000	0.0000	0.0000	0.0000
0.0000	1.0000	1.0000	-1.0000	0.0000	1.0000	0.0000	5.0000
0.0000	6.0000	5.0000	0.0000	-1.0000	0.0000	1.0000	10.0000

Intermediate Tables

Iteration 1

Entering variable: x_1 (most negative coefficient in the Z row: -36.0000). Leaving variable: s_2 (row 2, minimum ratio 1.6667).

Finding the Pivot

Z	x_1	x_2	x_3	x_4	s_1	s_2	b	P
1.0000	-36.0000	-30.0000	3.0000	4.0000	0.0000	0.0000	0.0000	-
0.0000	1.0000	1.0000	-1.0000	0.0000	1.0000	0.0000	5.0000	5.0000
0.0000	**6.0000**	5.0000	0.0000	-1.0000	0.0000	1.0000	10.0000	1.6667

Canonize the Pivot column with Pivot value

Z	x_1	x_2	x_3	x_4	s_1	s_2	b
1.0000	0.0000	0.0000	3.0000	-2.0000	0.0000	6.0000	60.0000
0.0000	0.0000	0.1667	-1.0000	0.1667	1.0000	-0.1667	3.3333
0.0000	1.0000	0.8333	0.0000	-0.1667	0.0000	0.1667	1.6667

Iteration 2

Entering variable: x_4 (most negative coefficient in the Z row: -2.0000). Leaving variable: s_1 (row 1, minimum ratio 20.0000).

Finding the Pivot

Z	x_1	x_2	x_3	x_4	s_1	s_2	b	P
1.0000	0.0000	0.0000	3.0000	-2.0000	0.0000	6.0000	60.0000	-
0.0000	0.0000	0.1667	-1.0000	**0.1667**	1.0000	-0.1667	3.3333	20.0000
0.0000	1.0000	0.8333	0.0000	-0.1667	0.0000	0.1667	1.6667	-

Canonize the Pivot column with Pivot value

Z	x_1	x_2	x_3	x_4	s_1	s_2	b
1.0000	0.0000	2.0000	-9.0000	0.0000	12.0000	4.0000	100.0000
0.0000	0.0000	1.0000	-6.0000	1.0000	6.0000	-1.0000	20.0000
0.0000	1.0000	1.0000	-1.0000	0.0000	1.0000	0.0000	5.0000

Iteration 3

The entering variable would be x_3 (coefficient -9.0000 in the Z row), but every entry of its column in the constraints is less than or equal to zero, so no row limits its growth: the problem is **unbounded**.

Z	x_1	x_2	x_3	x_4	s_1	s_2	b
1.0000	0.0000	2.0000	-9.0000	0.0000	12.0000	4.0000	100.0000
0.0000	0.0000	1.0000	-6.0000	1.0000	6.0000	-1.0000	20.0000
0.0000	1.0000	1.0000	-1.0000	0.0000	1.0000	0.0000	5.0000

Final Table

Z	x_1	x_2	x_3	x_4	s_1	s_2	b
1.0000	0.0000	2.0000	-9.0000	0.0000	12.0000	4.0000	100.0000
0.0000	0.0000	1.0000	-6.0000	1.0000	6.0000	-1.0000	20.0000
0.0000	1.0000	1.0000	-1.0000	0.0000	1.0000	0.0000	5.0000

Solution

Solution

The problem is **unbounded**: there is no finite optimal solution.

Unbounded problem

In iteration 3 the variable x_3 has an improving coefficient in the Z row (-9.0000), but all the entries of its column in the constraint rows are ≤ 0. Therefore the ratio test has no candidate row and x_3 can grow indefinitely without violating any constraint, improving Z without limit. The problem has no finite optimal solution.